ConvexHire

Bridging Multi-Agent Systems, RAG, and Explainable AI for Transparent Hiring

Final Year Project Report

Student Name Rahul Dev Banjara
University ID 2407061
Supervisor Ms. Prakriti Regmi
Reader Mr. Siman Giri
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Table of Contents

Acknowledgements ..... **vi**
Abstract **1**
1 Introduction **2**
1.1 Project Briefing 2
1.1.1 AI Aspect 2
1.2 Aims 2
1.3 Objectives 2
1.4 Artefact 2
1.4.1 FDD 2
1.5 Academic Question 3
2 Software Requirements Specification (SRS) **4**
2.1 Introduction 4
2.1.1 Purpose 4
2.1.2 Scope 4
2.1.3 Definitions, Acronyms, and Abbreviations 4
2.2 Overall Description 4
2.2.1 Product Perspective 4
2.2.2 User Classes and Characteristics 4
2.2.3 Design and Implementation Constraints 4
2.3 System Features (Subsystems) 4
2.3.1 Subsystem 1 Authentication, User Management and Company Management 4
2.3.2 Subsystem 2 Machine Learning based and Multi Agent Shortlisting . 4
2.3.3 Subsystem 3 Recruiter Chatbot with Agentic RAG 5
2.3.4 Subsystem 4 AI Resume Parsing and Job Fit Gap Analysis 5
2.3.5 Subsystem 5 Fine-Tuned ASR for Interview Transcription and Scoring 5
2.3.6 Subsystem 6 AI Job Drafting and Hybrid Semantic Search 5
2.3.7 Subsystem 7 AI Outreach Drafting and Interview Scheduling 5
2.4 Non-Functional Requirements 6
2.4.1 Performance Requirements 6
2.4.2 Security Requirements 6
2.4.3 Software Quality Attributes 6
2.5 External Interface Requirements 6
2.5.1 User Interfaces 6
2.5.2 Software Interfaces 6
2.5.3 Wireframes 6
2.6 Analysis Models 6
2.6.1 Data Model (ERD) 6
2.6.2 Behavioral Models 6
3 Literature Review **7**
3.1 AI in Talent Acquisition and Automated Screening 7
3.2 Resume Analysis, Semantic Matching, and Hybrid Retrieval 7
3.3 Multi-Agent and Multi-Persona LLM Architectures 8
3.4 Cost-Efficient AI Pipelines 8

3.5 Security and Privacy in AI Pipelines ... 8
3.6 Multilingual and Low-Resource Speech Recognition ... 9
3.7 Human-in-the-Loop AI for Consequential Decisions ... 9
3.8 Similar Systems ... 9
3.8.1 Nepal-Based Systems ... 9
3.8.2 Commercial Platforms ... 10
3.8.3 Research Prototypes ... 10
3.9 Gap Analysis ... 10
4 Project Methodology ... 12
4.1 Framework Justification (Kanban) ... 12
4.2 High-level Gantt Chart ... 12
5 Tools and Technologies ... 13
6 Artefact Designs ... 14
6.1 Subsystem 1 Authentication, User Management and Company Management ... 14
6.2 Subsystem 2 Machine Learning based and Multi Agent Shortlisting ... 18
6.3 Subsystem 3 Recruiter Chatbot with Agentic RAG ... 21
6.4 Subsystem 4 AI Resume Parsing and Job Fit Gap Analysis ... 23
6.5 Subsystem 5 Fine-Tuned ASR for Interview Transcription and Scoring ... 26
6.6 Subsystem 6 AI Job Drafting and Hybrid Semantic Search ... 29
6.7 Subsystem 7 AI Outreach Drafting and Interview Scheduling ... 32
6.8 Database Schema Diagram ... 35
6.9 Finetuning OpenAI Whisper ... 35
6.9.1 Integration ... 35
7 Test Plans ... 36
7.1 Introduction and QA Philosophy ... 36
7.2 Traditional Software Testing (Deterministic) ... 36
7.2.1 Unit and Integration Testing ... 36
7.2.2 Functional Black-Box Testing ... 36
7.3 AI Intelligence Benchmarking (Probabilistic) ... 36
7.3.1 R&D Benchmark OCR and Layout Parsing ... 36
7.3.2 Information Extraction (IE) and Structured Schema Validation ... 36
7.3.3 RAG Evaluation (RAGAS Framework) ... 36
7.3.4 ASR Performance Whisper Fine-tuning ... 36
7.4 Guardrails ... 36
7.4.1 Input Guardrail Prompt Injection Defense ... 37
7.4.2 Content Guardrail PII Redaction and Bias Mitigation ... 37
7.4.3 Output Guardrail Hallucination Prevention ... 37
8 Critical Evaluation ... 39
8.1 Final Report Reflection ... 39
8.2 Findings and System Evaluation ... 39
8.3 System Evaluation ... 39
8.4 Self-Reflection and Professional Development ... 39
9 Evidence of Project Management ... 40
9.1 Log Sheets ... 40
9.2 Gantt Chart ... 40
9.3 Trello Board ... 40
References ... 41

10 Appendices 43
10.1 Wireframes 43
10.2 Additional Diagrams 43

List of Figures

Figure 1 Feature Decomposition Diagram of ConvexHire 2
Figure 2 Entity-Relationship Diagram 6
Figure 3 High-level Gantt Chart 12
Figure 4 Authentication, User Management and Company Management - activity diagram 15
Figure 5 Authentication, User Management and Company Management - class diagram 16
Figure 6 Authentication, User Management and Company Management - sequence diagram 17
Figure 7 ML & Multi-Agent Shortlisting - activity diagram 19
Figure 8 ML & Multi-Agent Shortlisting - class diagram 20
Figure 9 ML & Multi-Agent Shortlisting - sequence diagram 21
Figure 10 Recruiter Chatbot with Agentic RAG - activity diagram 22
Figure 11 Recruiter Chatbot with Agentic RAG - class diagram 23
Figure 12 Recruiter Chatbot with Agentic RAG - sequence diagram ... 23
Figure 13 AI Resume Parsing and Job Fit Gap Analysis - activity diagram 24
Figure 14 AI Resume Parsing and Job Fit Gap Analysis - class diagram 25
Figure 15 AI Resume Parsing and Job Fit Gap Analysis - sequence diagram 26
Figure 16 Fine-Tuned ASR for Interview Transcription and Scoring - activity diagram 27
Figure 17 Fine-Tuned ASR for Interview Transcription and Scoring - class diagram 28
Figure 18 Fine-Tuned ASR for Interview Transcription and Scoring - sequence diagram 29
Figure 19 AI Job Drafting and Hybrid Semantic Search - activity diagram 30
Figure 20 AI Job Drafting and Hybrid Semantic Search - class diagram 31
Figure 21 AI Job Drafting and Hybrid Semantic Search - sequence diagram 32
Figure 22 AI Outreach Drafting and Interview Scheduling - activity diagram 33
Figure 23 AI Outreach Drafting and Interview Scheduling - class diagram 34
Figure 24 AI Outreach Drafting and Interview Scheduling - sequence diagram 34
Figure 25 PostgreSQL schema diagram 35
Figure 26 Detailed Gantt chart for the development lifecycle 40
Figure 27 Kanban board on Trello 40
Figure 28 Candidate dashboard wireframe 43

List of Tables

Table 1 Technology stack overview 13
Table 2 OCR and layout parsing comparative analysis 36

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Abstract

ConvexHire is an AI-powered recruitment platform that streamlines end-to-end hiring for both job seekers and employers. Candidates leverage AI-driven resume import, profile building, and resume gap analysis to strengthen applications, while discovering relevant roles through hybrid semantic-keyword search. Recruiters automate job drafting, candidate evaluation, and intelligent shortlisting, with a conversational AI assistant for talent search and salary research. Interview recordings are transcribed using a fine-tuned Whisper model for accurate assessment. All AI-generated actions-from outreach emails to job edits-require explicit human approval, ensuring transparency and recruiter control. The platform maintains real-time application tracking with visible AI scores and feedback, creating a fair, efficient hiring pipeline where artificial intelligence augments human decision-making rather than replacing it.

1 Introduction

1.1 Project Briefing

project briefing paragraph.

1.1.1 AI Aspect

AI aspect overview (NLP, ASR, Generative & Agentic AI).

1.1.1.1 Learning Justification

supervised learning + zero/few-shot reasoning justification.

1.1.1.2 Mathematics

cosine similarity and WER equations.

Vector Space Modeling - cosine similarity

$$\text{CosineSimilarity} = \frac{A \cdot B}{\|A\| \cdot \|B\|}$$

Word Error Rate (WER)

$$\text{WER} = \frac{S + D + I}{N}$$

where S = substitutions, D = deletions, I = insertions, and N = total reference words.

1.2 Aims

primary aim statement.

1.3 Objectives

bullet list of objectives.

1.4 Artefact

1.4.1 FDD

FDD figure + seven-subsystem decomposition.

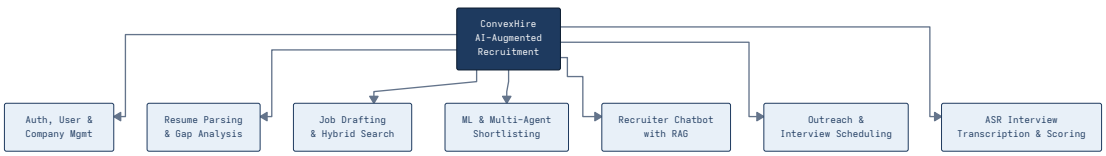


Figure 1: Feature Decomposition Diagram of ConvexHire

1.5 Academic Question

academic question statement.

2 Software Requirements Specification (SRS)

2.1 Introduction

2.1.1 Purpose

2.1.2 Scope

2.1.3 Definitions, Acronyms, and Abbreviations

MAS, RAG, NER, HITL, PII, ASR.

2.2 Overall Description

2.2.1 Product Perspective

2.2.2 User Classes and Characteristics

Candidate / Recruiter / Company Administrator.

2.2.3 Design and Implementation Constraints

regional language, multi-tenancy, observability.

2.3 System Features (Subsystems)

2.3.1 Subsystem 1 Authentication, User Management and Company Management

ConvexHire uses a dual-path authentication system supporting both local email-password registration and Google OAuth. Every signup requires email verification before the account becomes active. Role selection happens at registration time candidates sign up directly, while recruiters and company administrators must complete a company affiliation step (either registering a new company or requesting to join an existing one) before their verification email is dispatched. Upon login, the system issues a short-lived JWT access token and a long-lived refresh token stored in an httpOnly cookie to protect against XSS attacks. Role-based access control gates every protected endpoint, with distinct permissions for candidates, recruiters, and company admins. Recruiter join requests start in a pending state and require explicit admin approval before the user can access company resources.

2.3.2 Subsystem 2 Machine Learning based and Multi Agent Shortlisting

RandomForest ML bouncer pre-filter, top-20% LLM multi-agent evaluation, two-stage pipeline.

2.3.3 Subsystem 3 Recruiter Chatbot with Agentic RAG

intent classification, tool-use agent, semantic candidate search, job editing, salary research, conversation summarization, SSE streaming.

2.3.4 Subsystem 4 AI Resume Parsing and Job Fit Gap Analysis

PDF extraction, Llama Cloud/Gemini structured parsing, resume gap analysis, auto-patch with human approval.

2.3.5 Subsystem 5 Fine-Tuned ASR for Interview Transcription and Scoring

Whisper Small fine-tuning for Nepali-English code-switching, S3 audio pipeline, transcript evaluation, hybrid ranking.

2.3.6 Subsystem 6 AI Job Drafting and Hybrid Semantic Search

LangGraph job drafting agent, salary benchmarking, hybrid dense+sparse retrieval with RRF fusion.

2.3.7 Subsystem 7 AI Outreach Drafting and Interview Scheduling

AI email draft generation with human approval, Cal.com API integration, webhook-driven booking status.

2.4 Non-Functional Requirements

2.4.1 Performance Requirements

2.4.2 Security Requirements

2.4.3 Software Quality Attributes

2.5 External Interface Requirements

2.5.1 User Interfaces

2.5.2 Software Interfaces

2.5.3 Wireframes

2.6 Analysis Models

2.6.1 Data Model (ERD)

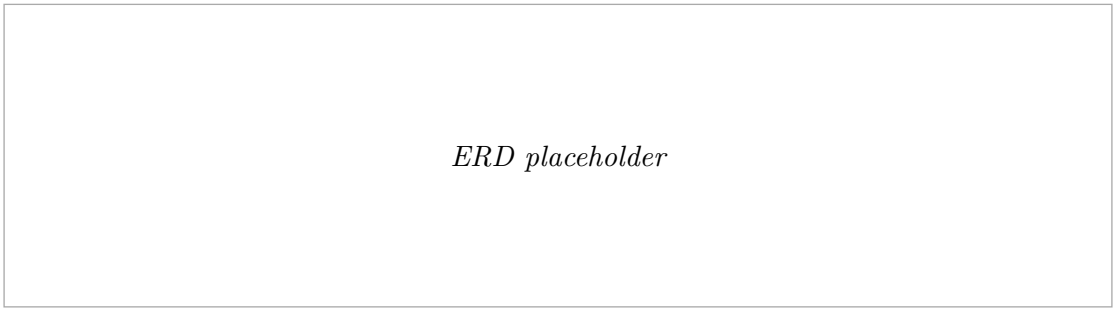


Figure 2: Entity-Relationship Diagram

2.6.2 Behavioral Models

LangGraph StateGraphs and sequence diagrams.

3 Literature Review

Recruitment has always been one of most labour-intensive processes in any organisation. With hundreds of applications arriving for single role, hiring teams face the impossible task of evaluating every candidate fairly and consistently within limited time. The rise of large language models, dense vector retrieval, and multi-agent AI systems have started to shift what is possible. This review examines the key research threads that underpin the design of ConvexHire, and analyses similar existing systems to identify the gaps this project aims to address.

3.1 AI in Talent Acquisition and Automated Screening

Automated candidate screening has moved rapidly from rule-based keyword matching to LLM-powered evaluation. Lo et al. [1] constructed a four-agent RAG-augmented pipeline for resume screening and showed the system could match the scoring quality of experienced HR professionals, something that earlier single-model approaches consistently failed to achieve at scale. Filatova et al. [13] demonstrated similar gains in a study of 64 real applicants, finding that an AI-assisted pipeline processed candidates in roughly half the time of an expert human recruiter, while still preserving human judgment at the final decision stage.

However the efficiency gains come with well-documented risks. Fabris et al. [10], in comprehensive survey of over 250 algorithmic hiring tools, found that even when protected attributes like gender and ethnicity are removed from the input, features such as university name and company brand act as proxies and reproduce the same discriminatory patterns through indirect correlations. They note that the EU AI Act now classifies hiring AI as high-risk, requiring meaningful transparency and auditability of the decision process. Peng et al. [9] compound this concern their AAAI study found that people in hybrid human-AI hiring settings show strong decision conformity, meaning they tend to follow the AI recommendation even when it is incorrect, meaning that poorly designed human-in-the-loop workflows are not necessarily safer than full automation.

3.2 Resume Analysis, Semantic Matching, and Hybrid Retrieval

Early resume processing relied on keyword matching, which failed to capture semantic equivalences across domain vocabulary. The move to transformer-based embeddings changed this significantly. Rosenberger et al. [2] developed CareerBERT, which maps both resumes and job descriptions into a shared embedding space using BERT and evaluates match quality through cosine similarity. Their system outperformed TF-IDF and vanilla BERT baselines in both application metrics and human expert evaluation, confirming that domain-adapted dense embeddings provide the right foundation for semantic candidate-job matching.

However CareerBERT has no exact-match fallback for precise technical terms, since searching for a specific certification or library name requires lexical precision that pure dense retrieval lacks. Hsu and Tzeng [3] address this directly with the DAT framework,

demonstrating that combining BM25 sparse retrieval with dense vector retrieval consistently outperforms either method alone across multiple retrieval benchmarks. The key insight is that semantic understanding and lexical precision are complementary capabilities, not competing ones, which directly motivates the hybrid Qdrant dense + BM25 design used in ConvexHire for both job discovery and recruiter candidate search.

3.3 Multi-Agent and Multi-Persona LLM Architectures

A foundational insight from recent LLM research is that multiple agents reasoning together and critiquing each other's outputs outperform any single agent on complex tasks. Du et al. [5], in work presented at ICLR, showed through controlled experiments that multi-agent debate, where agents propose answers independently, exchange critiques, and iterate across rounds, which substantially improved factual accuracy on arithmetic, reasoning, and biographical tasks. The mechanism is that disagreement surfaces errors which individual models consistently miss when reasoning alone.

Lo et al. [1] applied this to resume screening and found the multi-agent approach matched HR professional quality. Filatova et al. [13] showed that assigning distinct roles to specialist agents, specifically a technical evaluator and a culture-fit evaluator, produced richer, more trustworthy rationales than single-prompt scoring. Both papers also acknowledge the cost problem each additional critique iteration multiplies inference cost proportionally, and this remain unaddressed in the published work. This gap directly motivated the Judge Agent's confidence-threshold early-exit (≥70%, max two iterations) in ConvexHire.

3.4 Cost-Efficient AI Pipelines

Sending every resume to a large LLM is economically prohibitive at hiring scale. Ong et al. [4] formalise this, proving through linear programming that cascade routing, which directs easy rejects through a lightweight gate before the expensive model, is the optimal strategy under quality constraints. The RandomForest pre-filter in ConvexHire implements this principle three deterministic features (years-of-experience match, skills-overlap ratio, cosine similarity via fastembed) are used to calibrate a percentile threshold that eliminates approximately 80 percent of LLM calls entirely, with cost savings validated by the theoretical result in [4].

For the conversational AI layer, research on intent classification [11] confirms that rule-based classifiers handling the majority of high-confidence inputs at zero cost, with LLM fallback only for the ambiguous tail, is both practical and effective. This is consistent with the intent router design in ConvexHire, where keyword rules handle around 80 percent of recruiter messages for free.

3.5 Security and Privacy in AI Pipelines

Integrating LLMs into recruitment pipelines creates attack surfaces that traditional software does not have. Liu et al. [6] formalise and benchmark prompt injection attacks using the HouYi framework, demonstrating that attacker-controlled content embedded in a document can override system instructions in majority of tested LLM applications. In hiring context, resume bodies are the primary indirect injection vector a candidate

can embed malicious instructions into their CV text attempting to manipulate the shortlisting model. Liu et al. [14] extend this with formal taxonomy across 36 applications and explicitly recommend defence-in-depth because no single defence achieves robust protection on its own. Sharma et al. [7] separately show that hybrid rule-NER pipelines achieve 91.1% F1 for PII detection in unstructured documents, validating the precision of pattern-based redaction strategies. Together these findings motivated the stacked defence in ConvexHire NeMo Guardrails at every LLM entry point, plus PIIMiddleware redacting identifying information before any shortlisting model processes the resume.

3.6 Multilingual and Low-Resource Speech Recognition

Whisper is the current frontier of multilingual ASR, but its training data is heavily skewed toward high-resource languages and its zero-shot performance on low-resource languages like Nepali is substantially degraded. Ojha et al. [8] address this directly, constructing a comprehensive dataset from publicly available Nepali speech corpora and fine-tuning Whisper across multiple model sizes. Their results confirm that supervised fine-tuning on target-language data produces substantial WER reductions, with a smaller fine-tuned model consistently outperforming larger zero-shot Whisper on Nepali. They also note that code-switched speech (English mixed with Nepali) was not systematically addressed, which is a remaining open problem. Whisper-LM [15] further shows that language-model rescoring can reduce WER additionally, though at the cost of inference latency that is unsuitable for real-time interview transcription workflows.

3.7 Human-in-the-Loop AI for Consequential Decisions

The empirical case for human oversight in high-stakes AI decisions is well established. Peng et al. [9] show that AI assistance improves human accuracy in hybrid hiring decisions when the AI model is well-designed, but that conformity effects can cause humans to accept incorrect recommendations without questioning them, especially when only a score is shown rather than the underlying reasoning. Filatova et al. [13] show that keeping a human final decision-maker in an AI hiring pipeline preserves accountability while still delivering meaningful efficiency gains compared to manual review. Both studies argue that transparency into AI reasoning is not a nice-to-have but a necessary condition for human oversight to function as intended, which directly informed ConvexHire's design of surfacing structured evidence alongside every shortlisting recommendation.

3.8 Similar Systems

3.8.1 Nepal-Based Systems

The most directly comparable Nepal-origin system found in the literature is an open-source AI-Powered Job Portal (Soum-ik/AI-Powerd-Job-Portal on GitHub) [S5]. It allows recruiters to generate job descriptions using a generative AI model and provides role-based access for candidates and recruiters. However, it offers no candidate screening, no semantic search, no interview transcription, and no multi-agent evaluation. It covers approximately one of the fifteen AI components present in ConvexHire, which

confirms that no Nepal-built system at time of writing addresses the full AI-assisted hiring funnel.

3.8.2 Commercial Platforms

Eightfold.ai [S1] is the most technically sophisticated commercial platform available, built on deep learning models trained across 1.6 billion career profiles with inferred skill taxonomies. It uses embedding-based semantic matching and has recently added agentic capabilities for autonomous task execution. Despite this technical depth, Eightfold functions as a closed-loop system for recruiters candidates have no visibility into why they were ranked, and the decision logic is not auditable by the people it affects [10]. There is no documented multi-agent critique loop, no Nepali ASR, and no evidence of prompt injection defence.

HireVue [S3] processes video interviews at scale using NLP models trained on over 70 million assessments, scoring responses against competency frameworks validated by IO psychologists. Its scoring uses single model rather than multi-persona critique, its ASR pipeline targets English-dominant markets with no documented support for low-resource South Asian languages, and candidates receive no feedback on their performance after assessment.

LinkedIn Recruiter [S4] (2024 release) introduced natural-language candidate search and AI-drafted outreach, reporting meaningful improvements in candidate response rates. However its candidate pool is limited to LinkedIn members, it offers no multi-agent shortlisting or interview evaluation, and no HITL confirmation exists at the per-candidate decision level.

3.8.3 Research Prototypes

Filatova et al. [13] describe an academic multi-agent hiring prototype with HITL validation, evaluated on 64 real applicants and achieving near-50% efficiency gain over human expert reviewers. It is the closest architectural analogue to ConvexHire in the academic literature. However it lacks an ML pre-filter (every candidate goes to the full LLM pipeline, which is prohibitive at scale), has no prompt injection defence, no PII redaction, and no semantic search for candidate discovery.

3.9 Gap Analysis

Three primary gaps emerge clearly from this review.

The first is the *explainability gap*. Commercial systems like Eightfold, LinkedIn Recruiter, and HireVue produce scores without justification that is visible to candidates or auditable by recruiters. Fabris et al. [10] identify this as both an ethical problem and an emerging legal requirement under EU AI regulation. Lo et al. [1] propose evidence-linked multi-agent evaluation as the structural solution. ConvexHire addresses this by requiring every shortlisting agent to produce structured strengths, concerns, and cited evidence, surfacing these to both recruiter and candidate, and embedding specific AI feedback in rejection emails.

The second is the *candidate feedback gap*. Tools like Teal and Jobscan give candidates generic improvement advice disconnected from any specific employer's actual evaluation

criteria. Paradox engages candidates conversationally at the top of funnel but evaluates nothing of substance. No system reviewed here combines in-context gap analysis against a real job description with constraint-safe AI resume patching (Frozen Facts) and structured post-shortlist feedback in one integrated candidate flow.

The third is the *integration gap*. No single platform reviewed integrates semantic job search, ML-gated multi-agent shortlisting, low-resource ASR for interview transcription, AI outreach drafting, and a recruiter AI assistant into one coherent pipeline. This gap is specially pronounced for non-English linguistic contexts, since no commercial or academic system reviewed addresses Nepali or code-switched interview transcription [S5], which is the specific challenge that motivated the Whisper fine-tuning work in ConvexHire.

4 Project Methodology

4.1 Framework Justification (Kanban)

4.2 High-level Gantt Chart

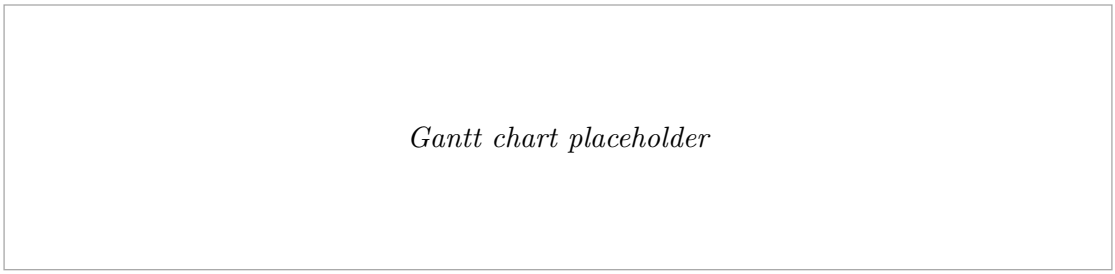


Figure 3: High-level Gantt Chart

5 Tools and Technologies

Category	**Tools**
Frontend Framework	Next.js (App Router), React, TypeScript, SCSS
Backend Framework	FastAPI, Pydantic
Database	PostgreSQL, SQLAlchemy, Alembic (migrations)
Vector / Cache	Qdrant (dense + sparse), fastembed, Redis
AI Orchestration	LangChain, LangGraph
LLM Providers	Groq, AWS Bedrock, Google Gemini
Web Search / Retrieval	Tavily
Document Parsing	Llama Cloud
ASR	Fine-tuned Whisper Small
ML	scikit-learn (RandomForest bouncer)
Guardrails / Safety	NVIDIA NeMo Guardrails, LangChain `PIIMiddleware`
Observability	LangSmith
AWS SDK	Boto3
Media Storage	Amazon S3
Email / Scheduling	aiosmtplib, Cal.com API
PDF / Templating	WeasyPrint, Jinja2
Rate Limiting	fastapi-limiter
Containerization	Docker
Deployment	Render, Vercel
IDE	Visual Studio Code, Cursor
Version Control	Git
Testing	PyTest
Package Manager	pnpm, uv

Table 1: Technology stack overview

justification paragraph for the chosen stack.

6 Artefact Designs

6.1 Subsystem 1 Authentication, User Management and Company Management

The activity diagram for authentication and user management flow is shown below. Users can sign up locally or via Google OAuth. All new accounts require email verification. Candidates are created immediately after verification, while recruiters must either register a new company or request to join an existing one before their account is activated. Recruiter join requests remain pending until approved by a company admin. Login issues a short-lived JWT access token and a long-lived refresh token stored in an httpOnly cookie. The refresh token is rotated on every use and can be revoked by changing the password. Profile management includes updating personal details, uploading an avatar, changing the password, or soft-deleting the account. Company administrators can update company profiles, manage recruiters, and approve or reject pending join requests.

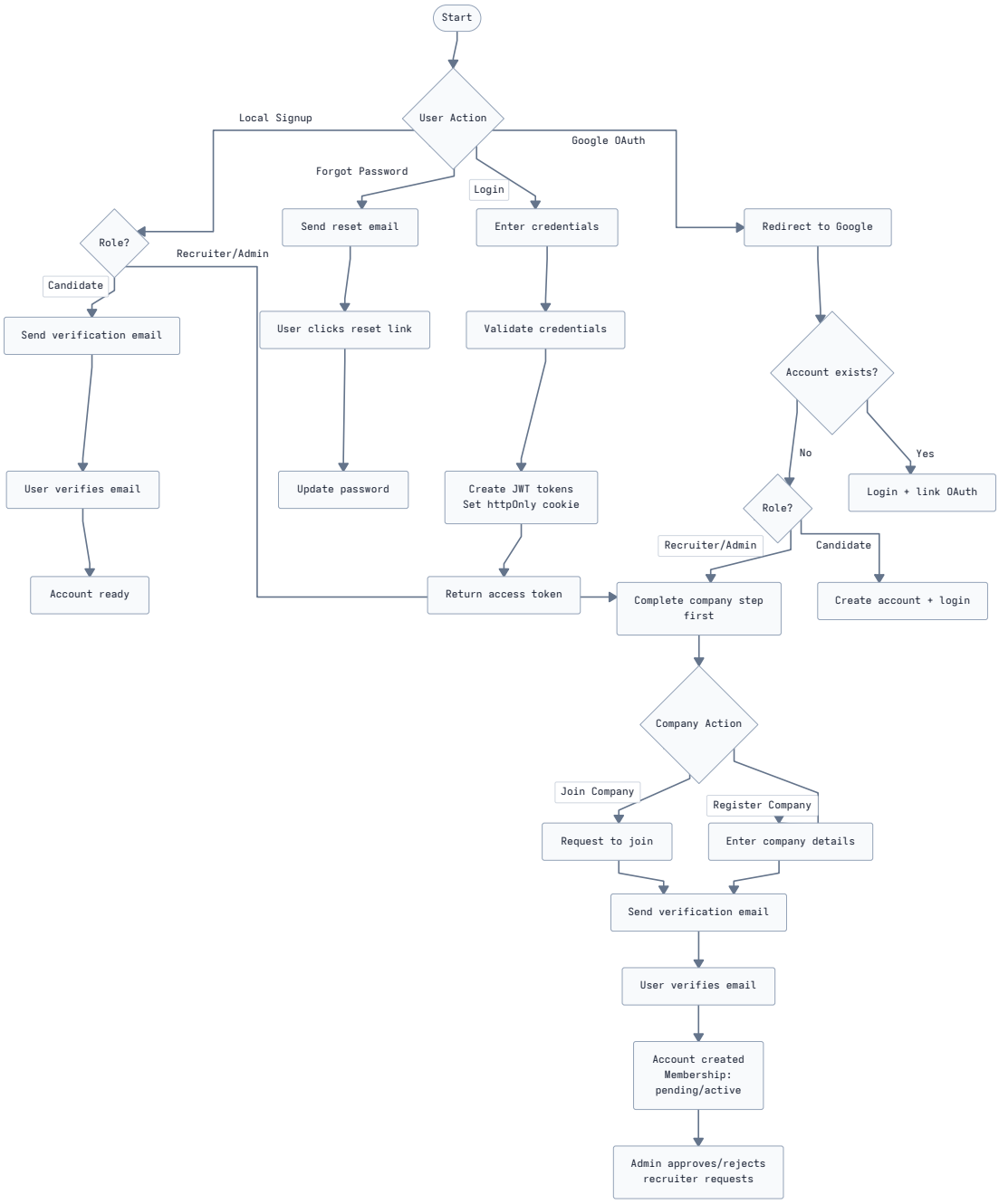


Figure 4: Authentication, User Management and Company Management - activity diagram

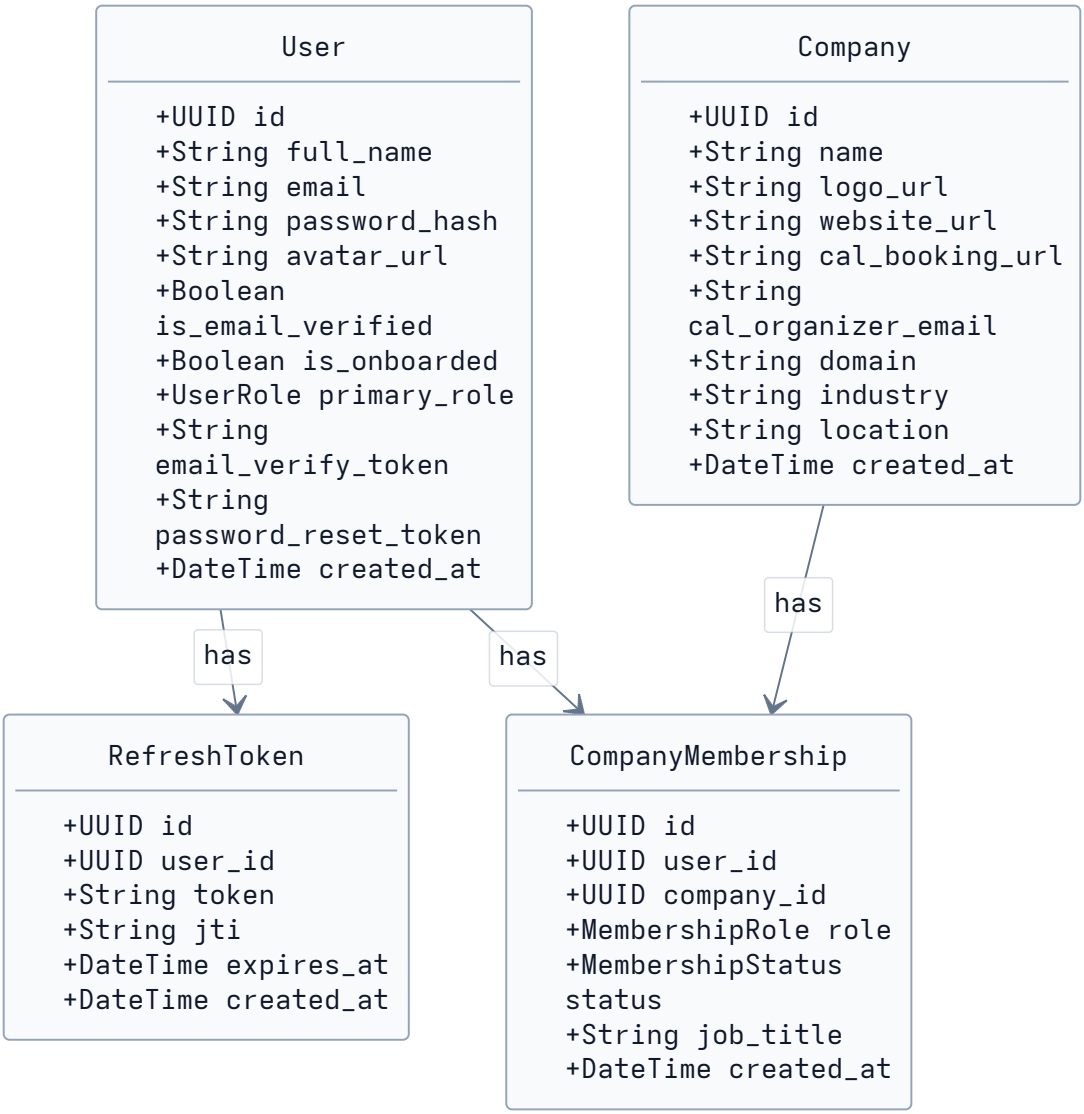


Figure 5: Authentication, User Management and Company Management - class diagram

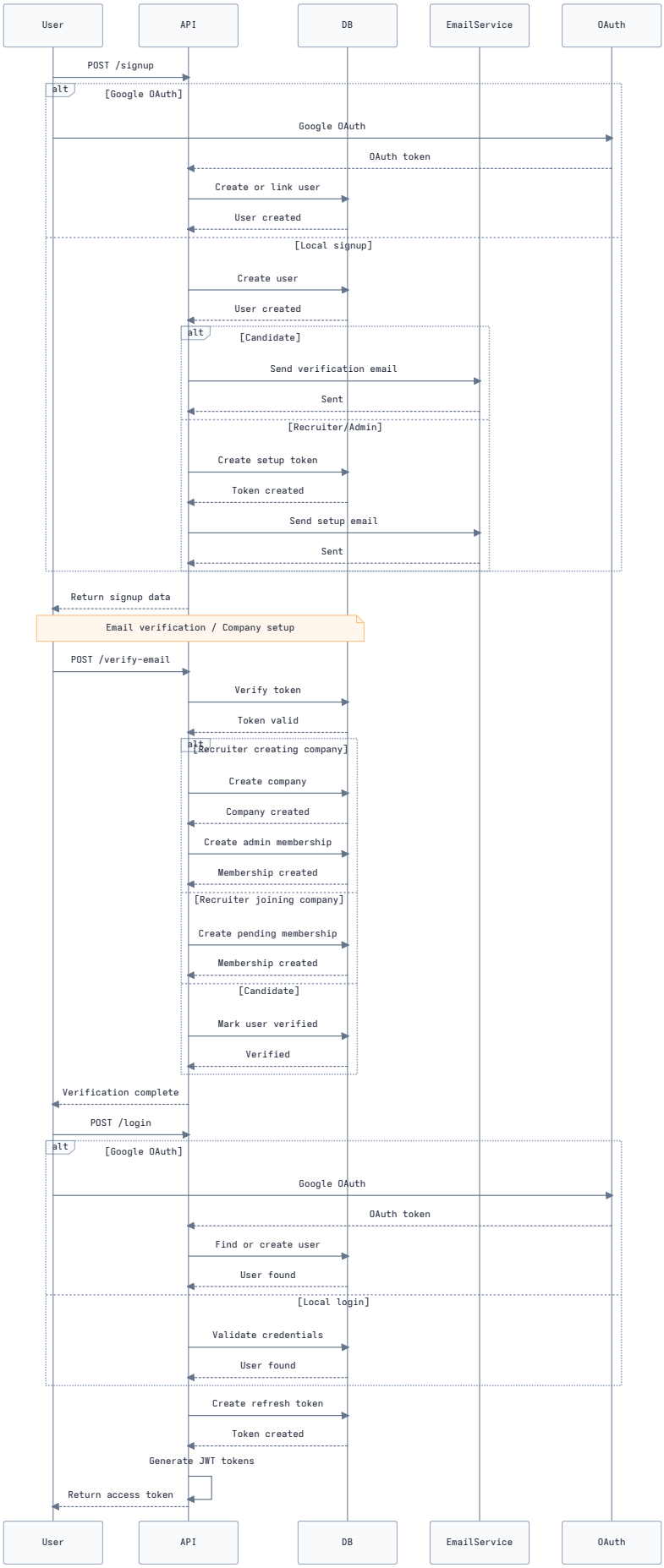


Figure 6: Authentication, User Management and Company Management - sequence diagram

6.2 Subsystem 2 Machine Learning based and Multi Agent Shortlisting

RandomForest feature extraction diagram + two-stage pipeline sequence + LangGraph StateGraph.

To keep shortlisting fair, the system removes personal details from resumes before AI reads them - names, emails, phone numbers, and links are stripped out or redacted. The AI is also told clearly to judge only on skills and experience, ignoring age, gender, nationality, or school reputation. More details are in the **PII Redaction and Bias Mitigation** section under security testing.

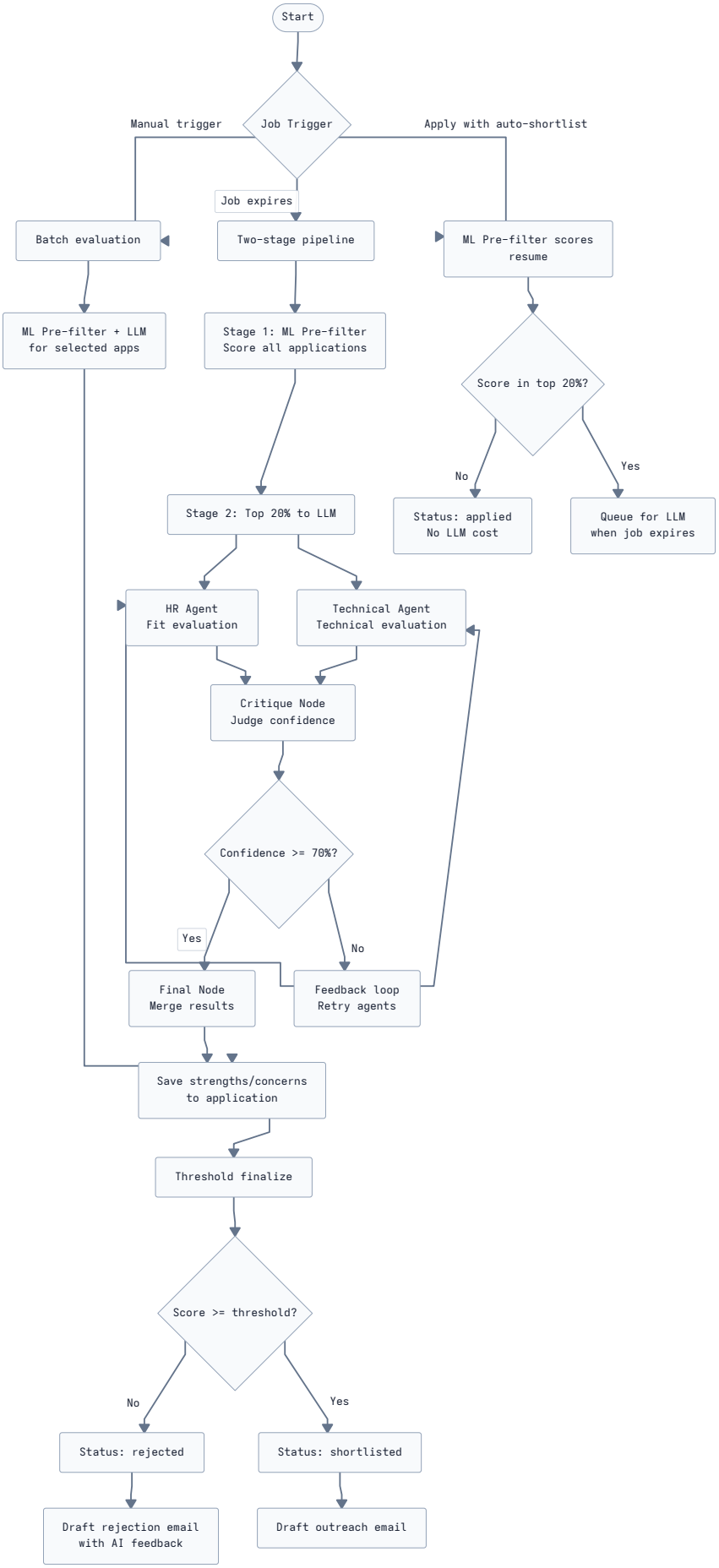


Figure 7: ML & Multi-Agent Shortlisting - activity diagram

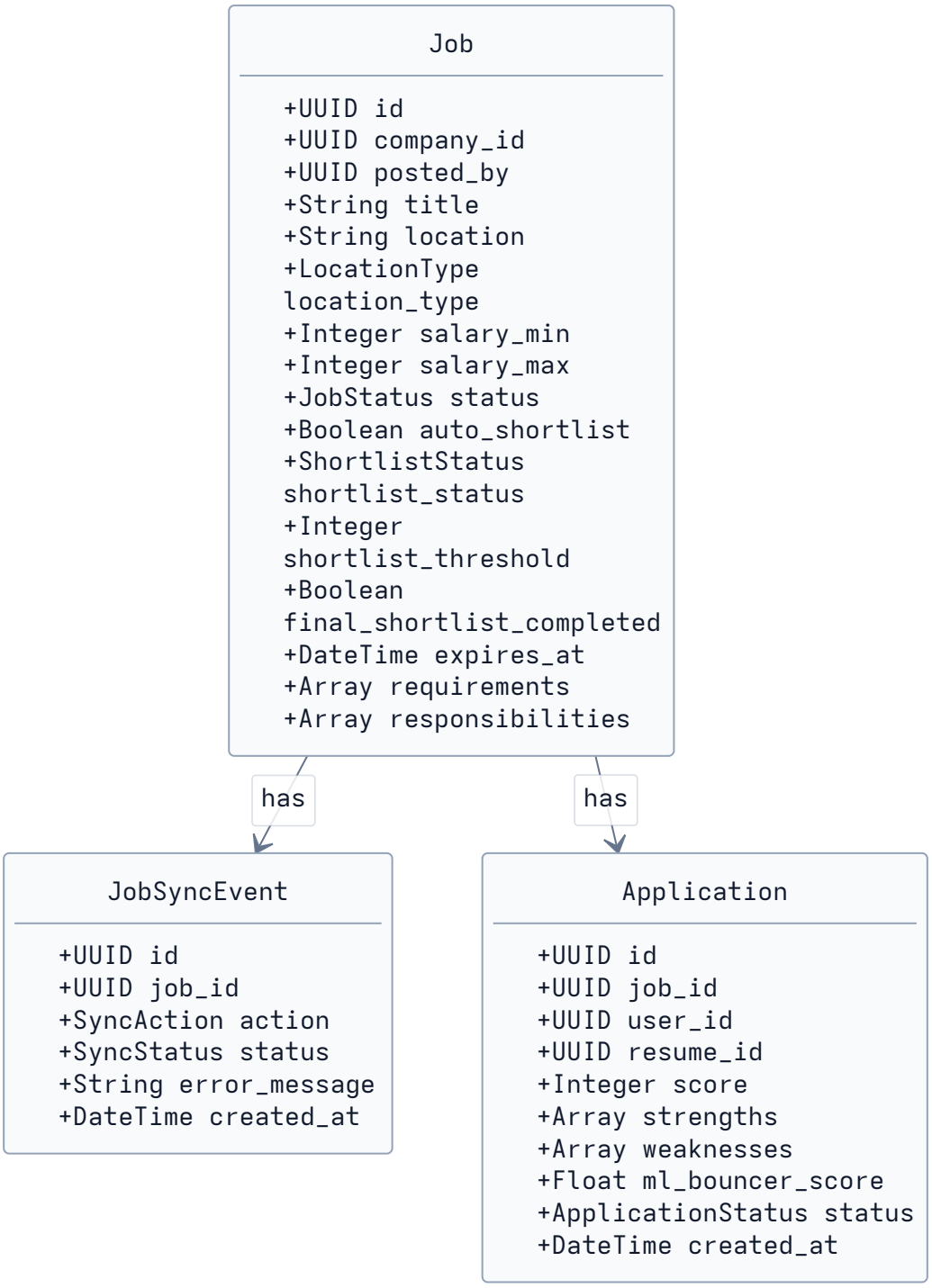


Figure 8: ML & Multi-Agent Shortlisting - class diagram

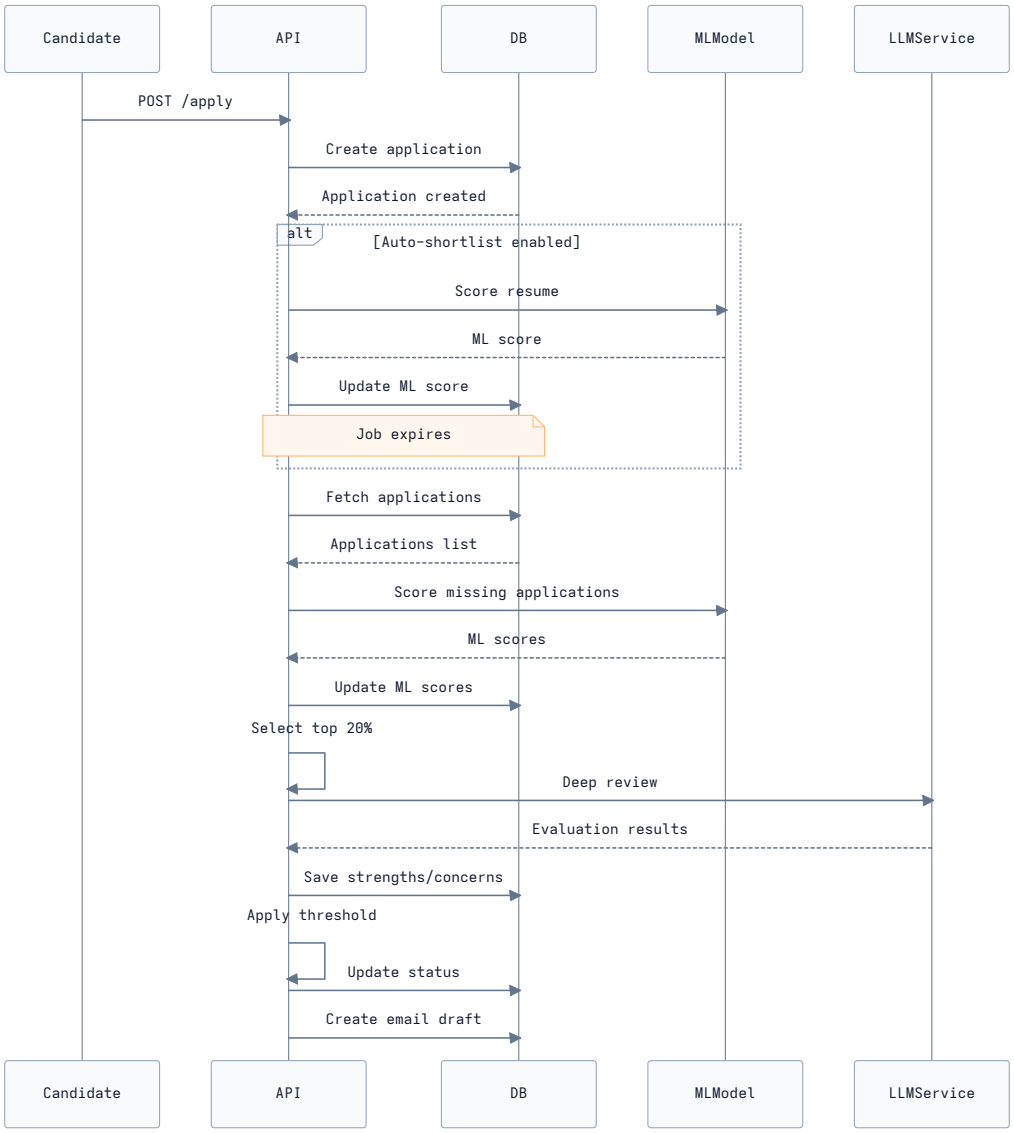


Figure 9: ML & Multi-Agent Shortlisting - sequence diagram

6.3 Subsystem 3 Recruiter Chatbot with Agentic RAG

intent routing diagram + tool-use sequence + SSE streaming architecture.

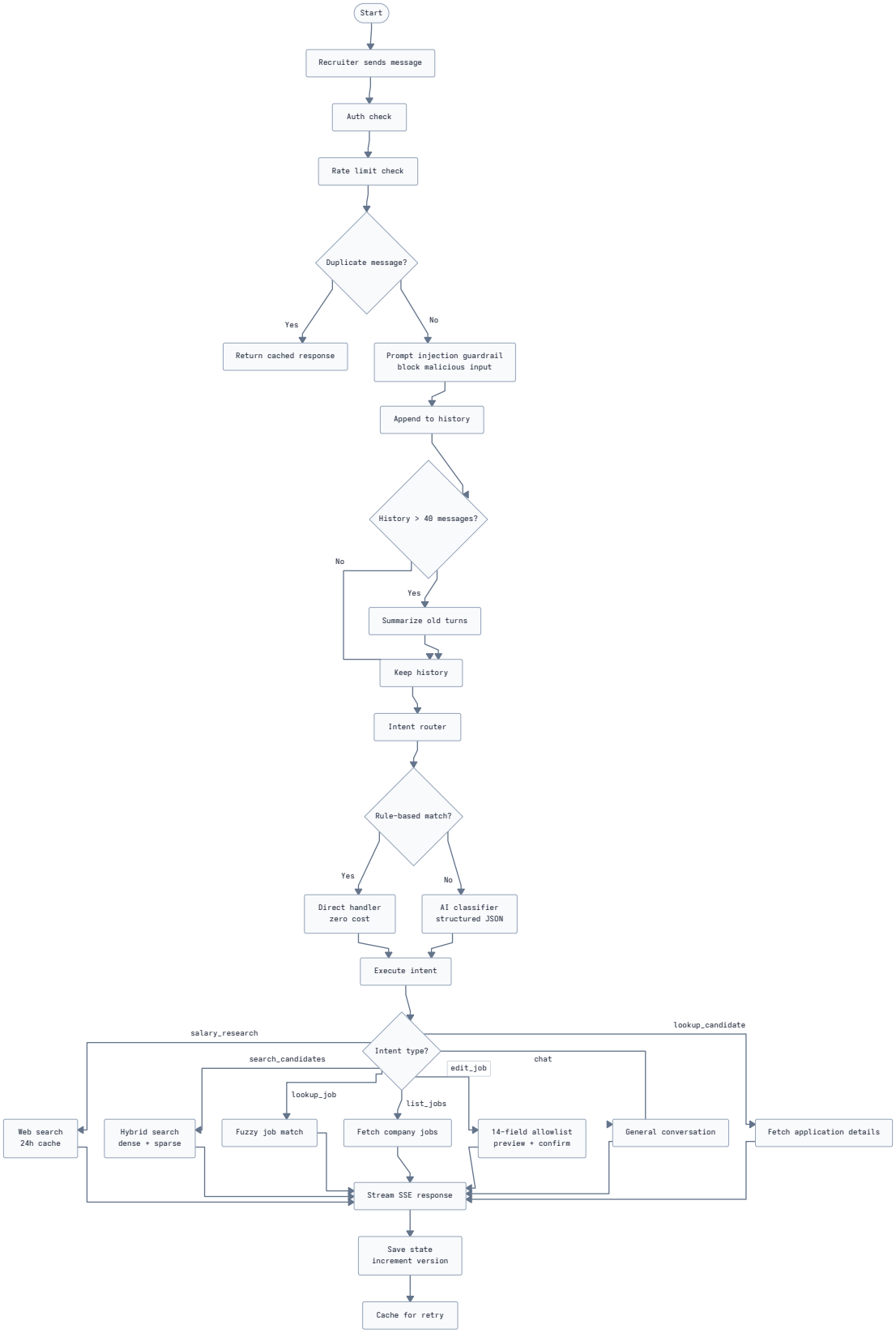


Figure 10: Recruiter Chatbot with Agentic RAG - activity diagram



Figure 11: Recruiter Chatbot with Agentic RAG - class diagram

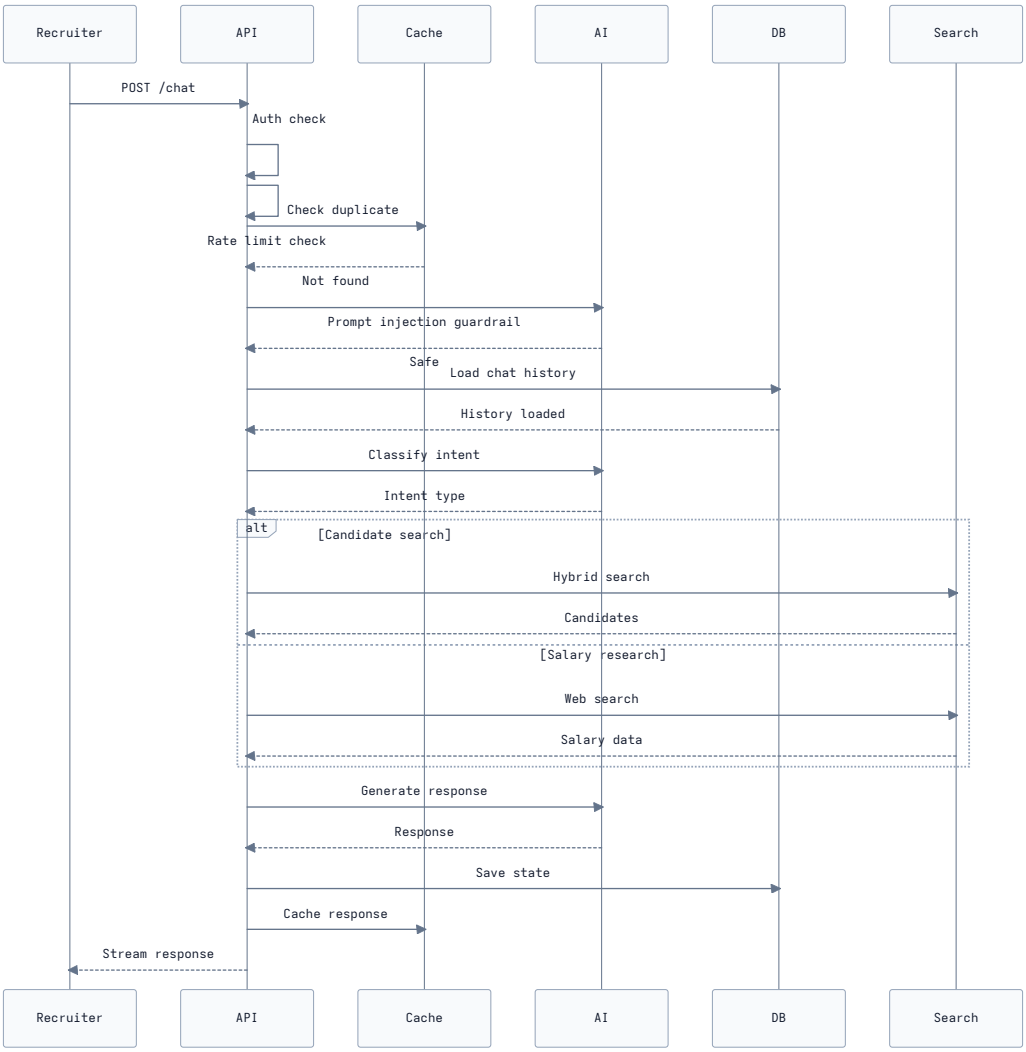


Figure 12: Recruiter Chatbot with Agentic RAG - sequence diagram

6.4 Subsystem 4 AI Resume Parsing and Job Fit Gap Analysis

PDF extraction pipeline + gap analysis activity + patch approval sequence.

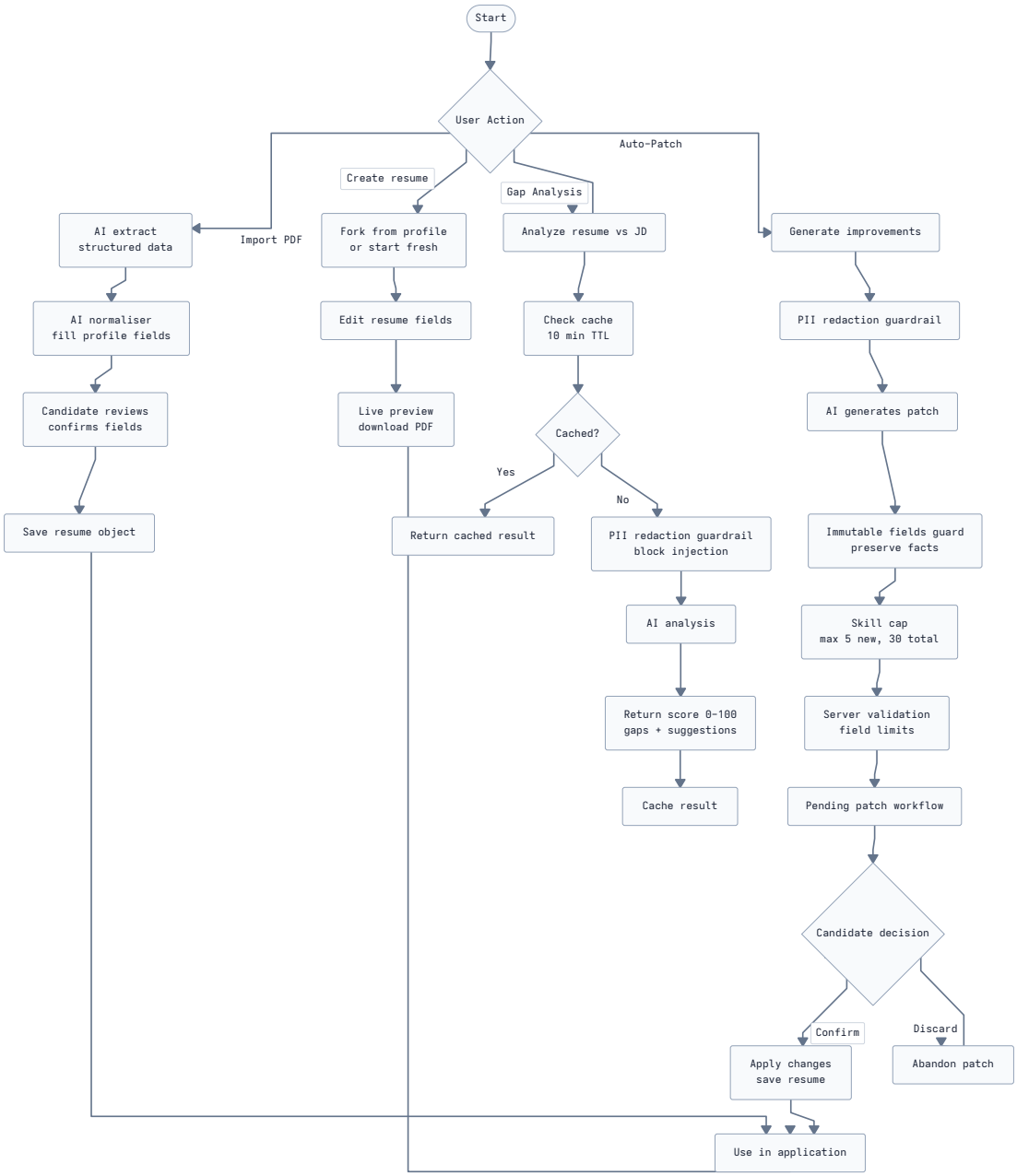


Figure 13: AI Resume Parsing and Job Fit Gap Analysis - activity diagram

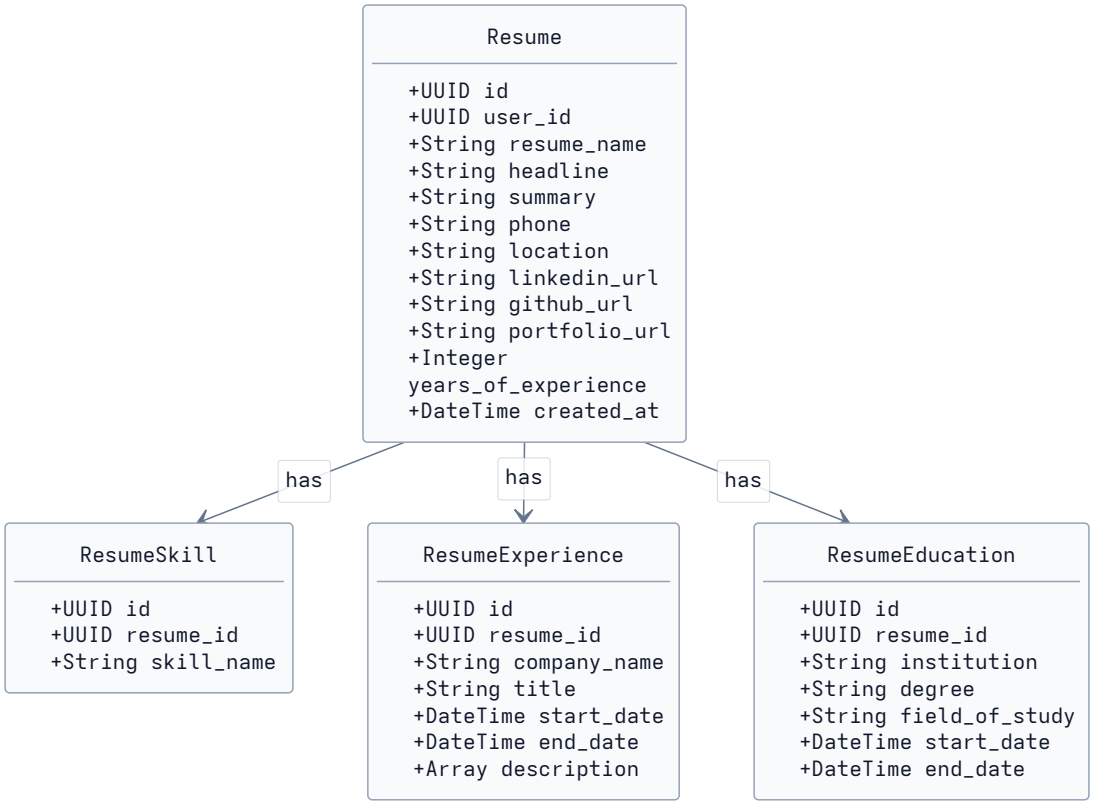


Figure 14: AI Resume Parsing and Job Fit Gap Analysis - class diagram

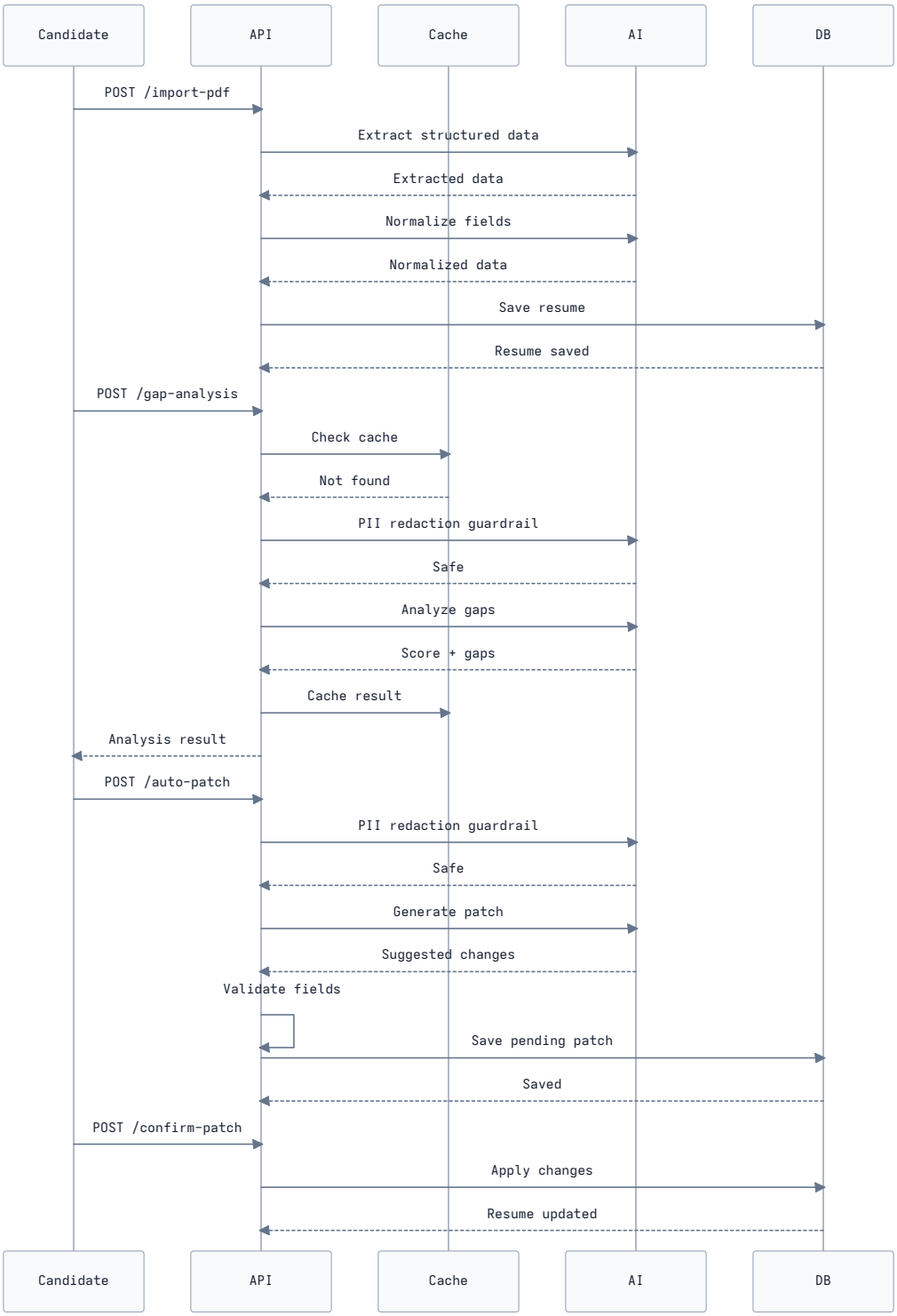


Figure 15: AI Resume Parsing and Job Fit Gap Analysis - sequence diagram

6.5 Subsystem 5 Fine-Tuned ASR for Interview Transcription and Scoring

Whisper fine-tuning curves + S3 audio pipeline + transcript evaluation sequence.

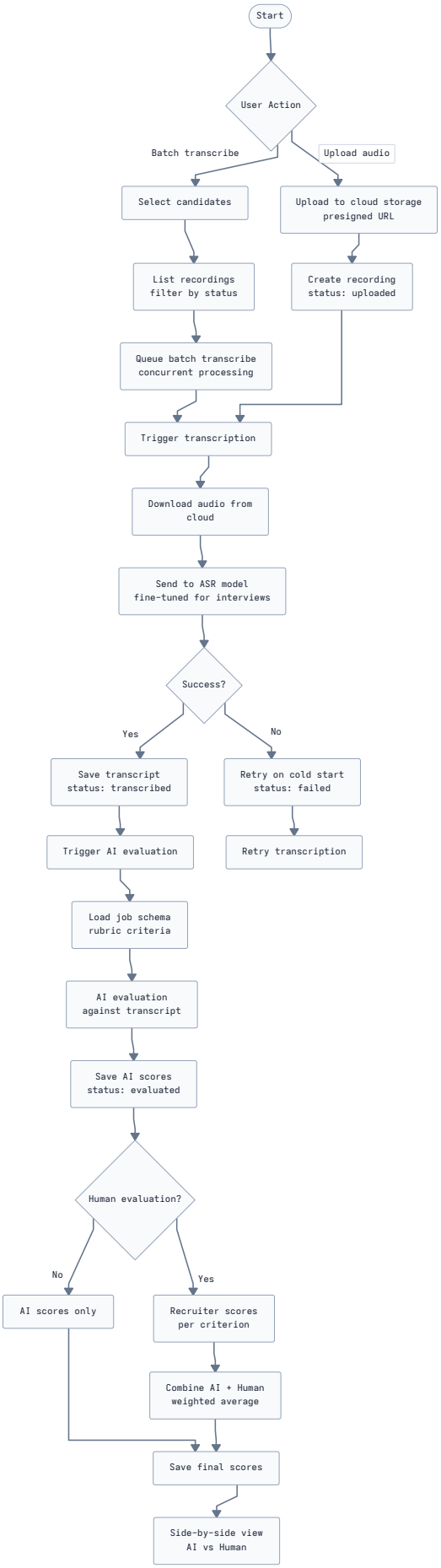


Figure 16: Fine-Tuned ASR for Interview Transcription and Scoring - activity diagram

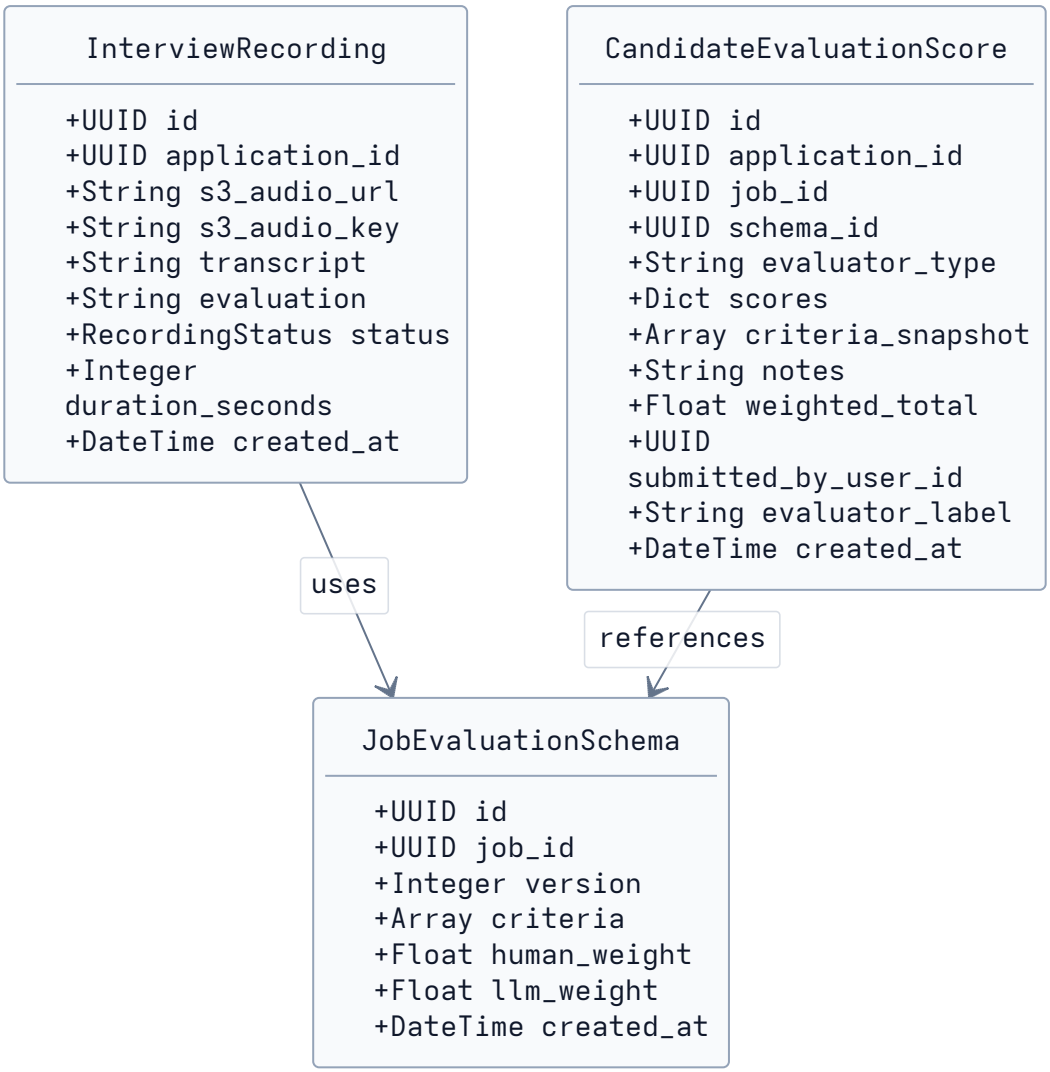


Figure 17: Fine-Tuned ASR for Interview Transcription and Scoring - class diagram

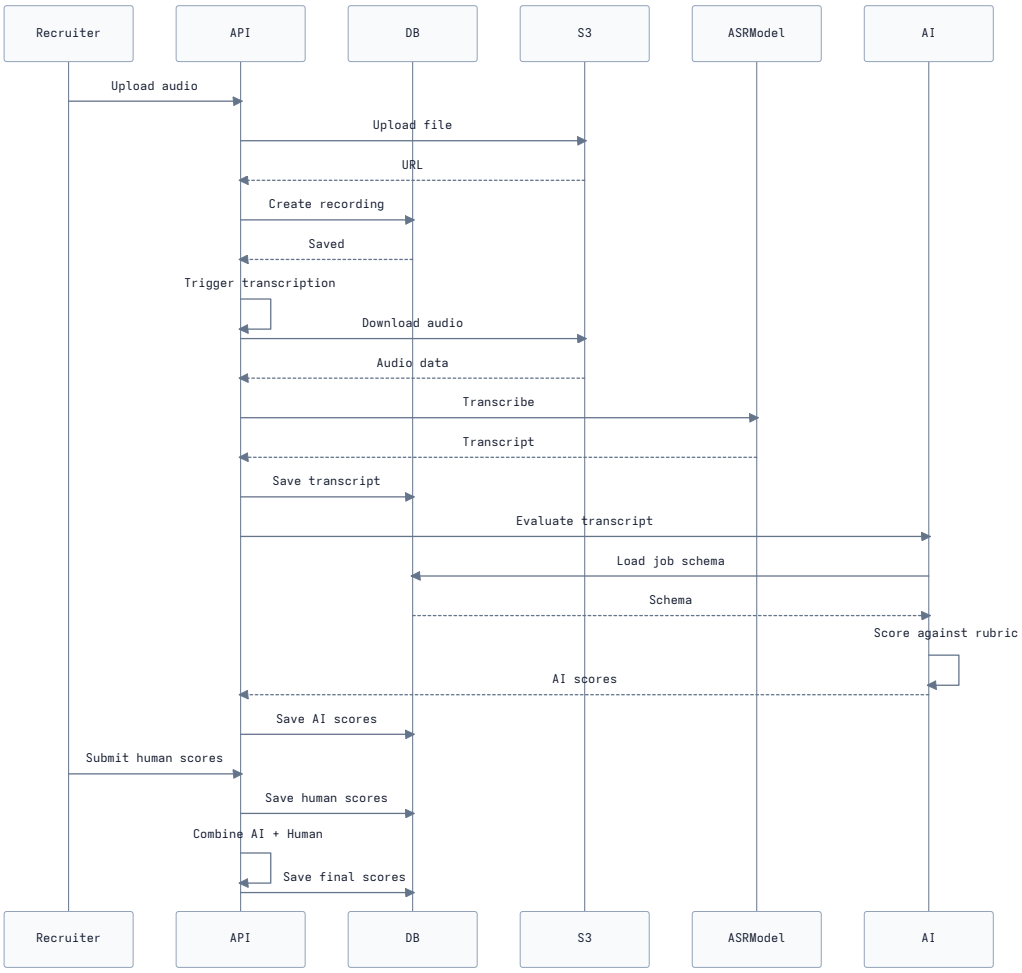


Figure 18: Fine-Tuned ASR for Interview Transcription and Scoring - sequence diagram

6.6 Subsystem 6 AI Job Drafting and Hybrid Semantic Search

LangGraph drafting state graph + hybrid search architecture (dense + sparse + RRF).

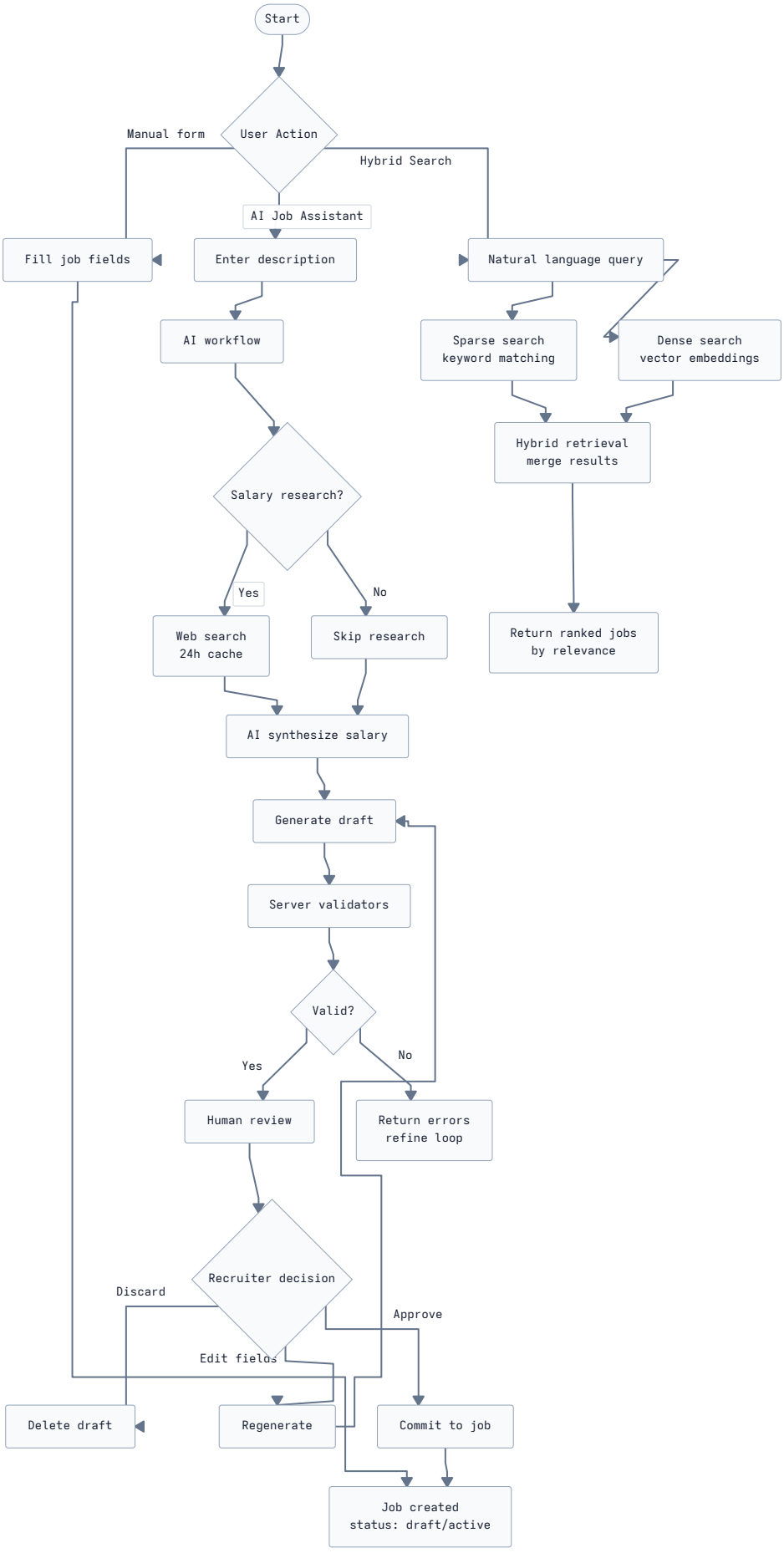


Figure 19: AI Job Drafting and Hybrid Semantic Search - activity diagram

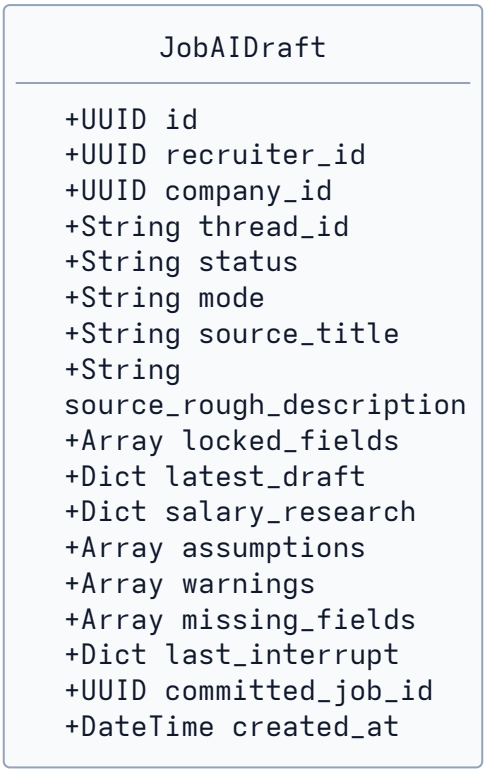


Figure 20: AI Job Drafting and Hybrid Semantic Search - class diagram

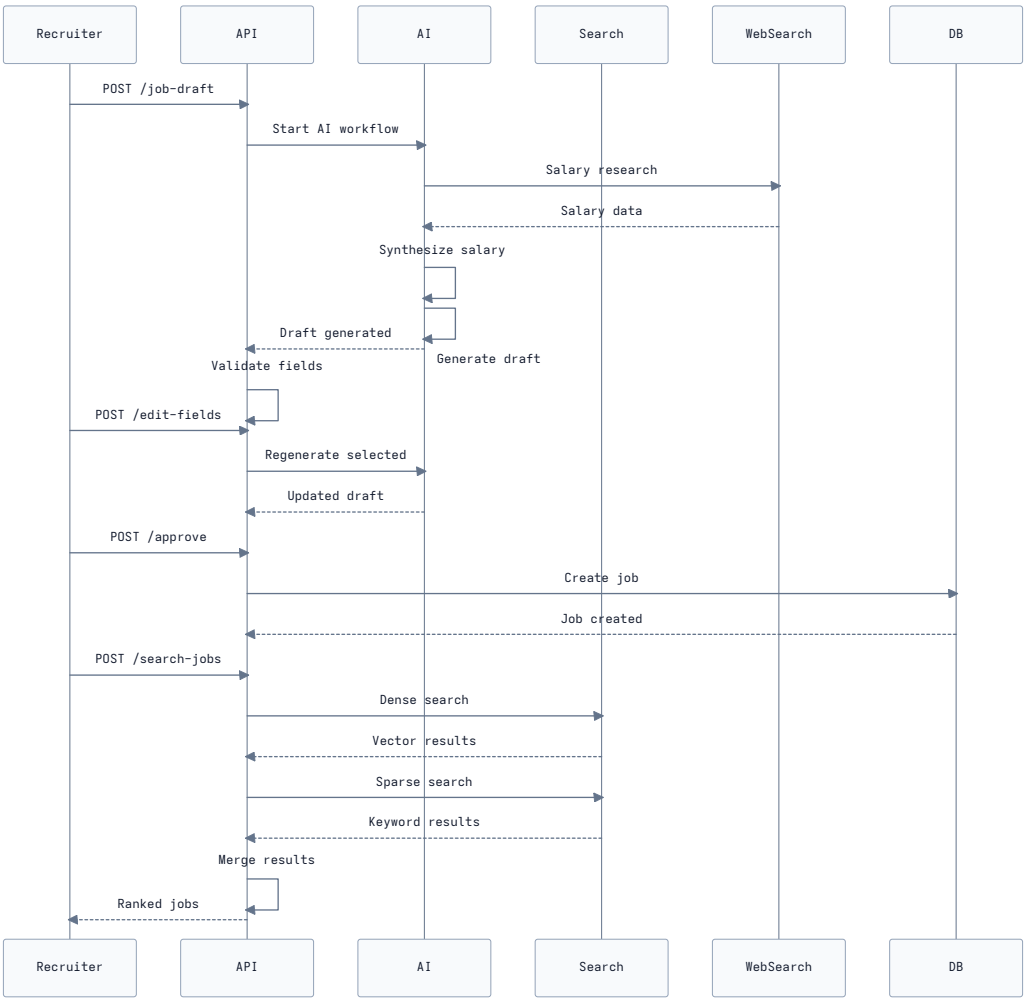


Figure 21: AI Job Drafting and Hybrid Semantic Search - sequence diagram

6.7 Subsystem 7 AI Outreach Drafting and Interview Scheduling

email draft approval activity + Cal.com webhook sequence + outreach class diagram.

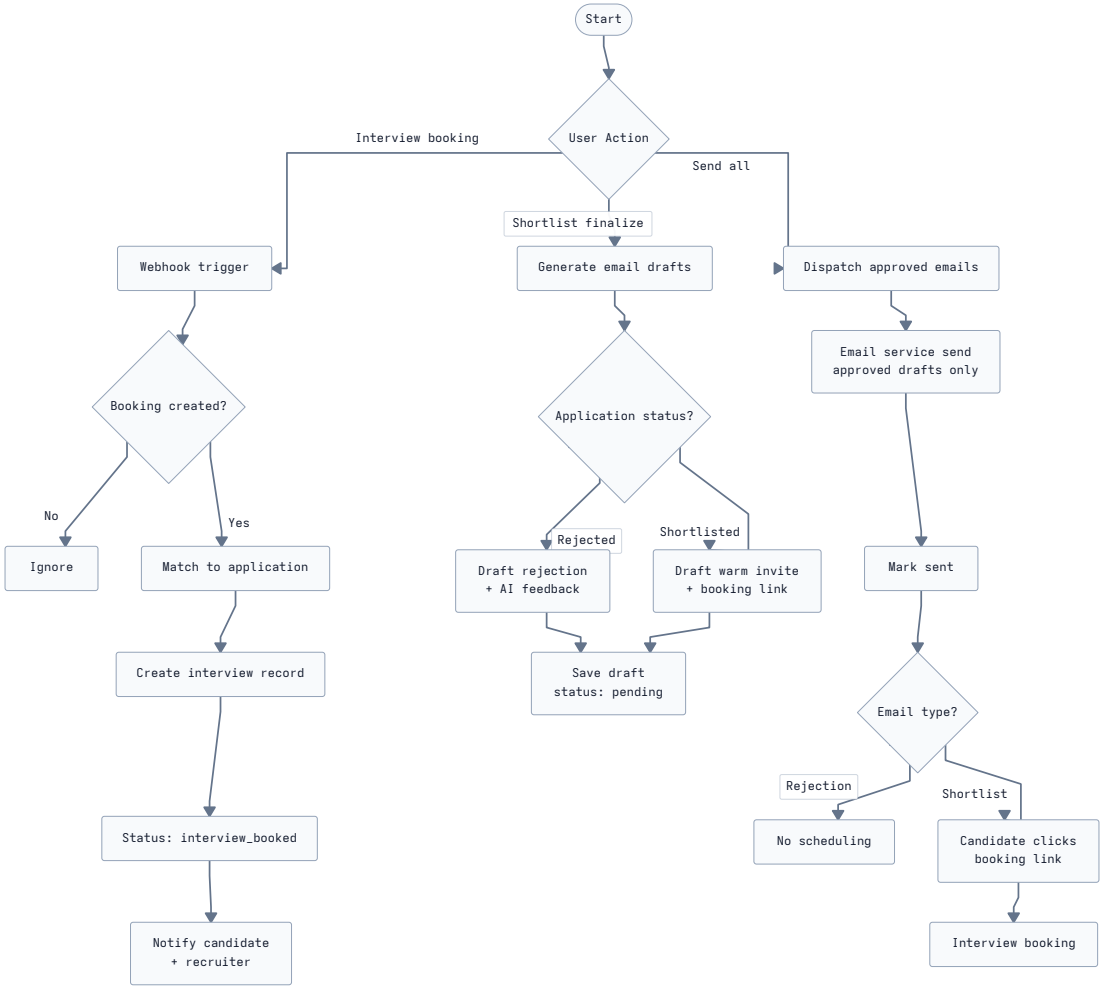


Figure 22: AI Outreach Drafting and Interview Scheduling - activity diagram

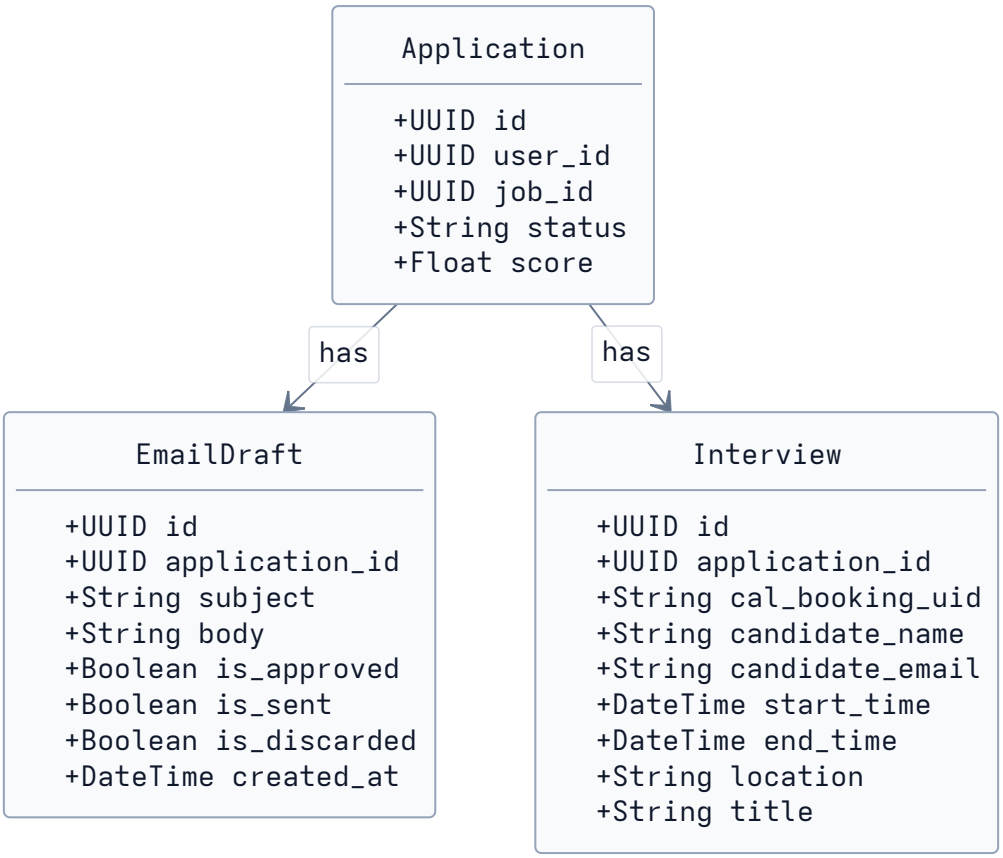


Figure 23: AI Outreach Drafting and Interview Scheduling - class diagram

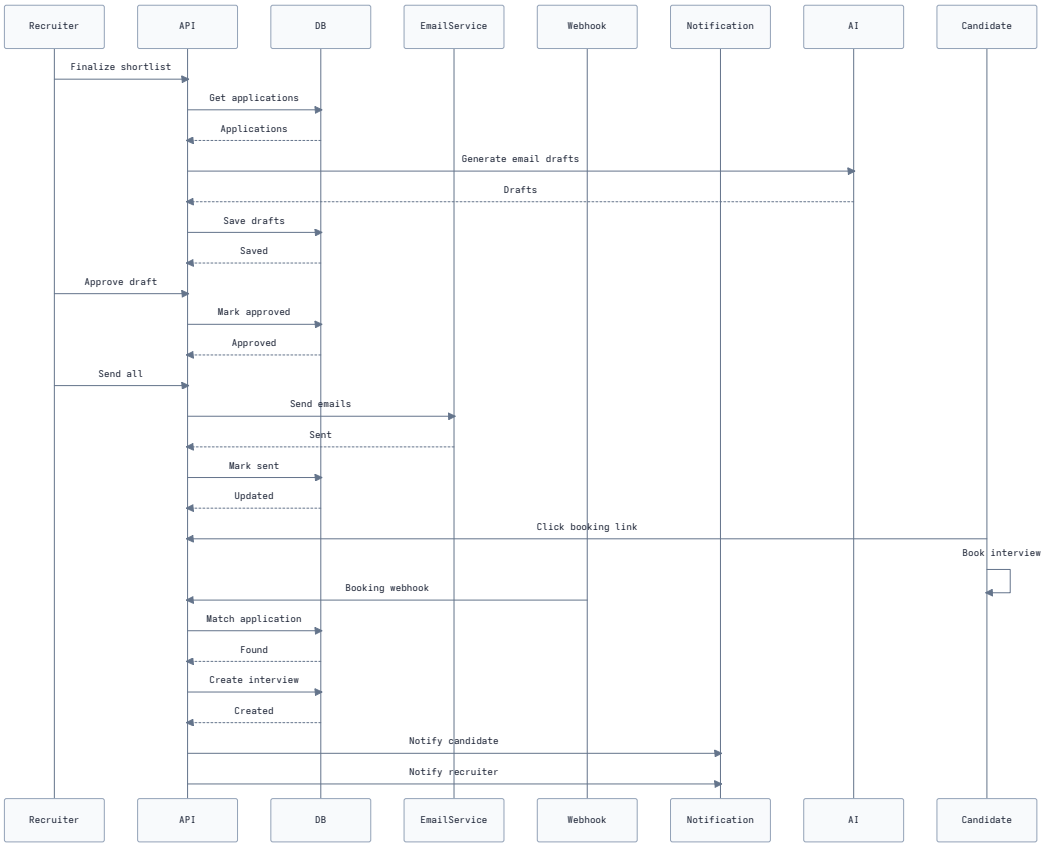


Figure 24: AI Outreach Drafting and Interview Scheduling - sequence diagram

6.8 Database Schema Diagram



Figure 25: PostgreSQL schema diagram

6.9 Finetuning OpenAI Whisper

synthetic data, crowdsourcing, storage.

6.9.1 Integration

decoupled FastAPI service for the ASR pipeline.

7 Test Plans

7.1 Introduction and QA Philosophy

deterministic vs probabilistic dual-track QA.

7.2 Traditional Software Testing (Deterministic)

7.2.1 Unit and Integration Testing

7.2.2 Functional Black-Box Testing

7.3 AI Intelligence Benchmarking (Probabilistic)

7.3.1 R&D Benchmark OCR and Layout Parsing

Metric	**Docling (IBM)**	**Chandra OCR**	**Llama Parse**	**Tesseract**
Layout Accuracy	95%	92%	90%	30%
LLM Compatibility	Markdown	JSON	Markdown	Raw Text
CPU Latency	Fast	Very Slow	N/A (Cloud)	Instant
Data Privacy	Local (Safe)	Local	Cloud	Local

Table 2: OCR and layout parsing comparative analysis

7.3.2 Information Extraction (IE) and Structured Schema Validation

ground truth, hallucination rate, omission rate.

7.3.3 RAG Evaluation (RAGAS Framework)

faithfulness, answer relevance, precision@K.

7.3.4 ASR Performance Whisper Fine-tuning

WER benchmark.

7.4 Guardrails

Guardrails are runtime checks that sit between the user, the language model, and the rest of the system. They decide what the model is allowed to read, what it is allowed to say, and what shape its answer must take. ConvexHire uses three layers of guardrails - one at the input, one over the content the model handles, and one over the output it produces - so that a single weak point cannot cause unfair, unsafe, or fabricated behaviour to reach the recruiter or candidate.

7.4.1 Input Guardrail Prompt Injection Defense

Resumes are untrusted documents. A candidate can paste hidden instructions into the body of their CV in an attempt to hijack the shortlisting agent - telling it to ignore the recruiter's rules, to give every applicant a perfect score, or to leak the system prompt. To stop this, ConvexHire wraps the shortlisting and recruiter chat models with NVIDIA NeMo Guardrails, configured with a `self_check_input` rail. Before the resume ever reaches the scoring model, a separate classifier LLM reads the incoming text and decides whether it is a normal recruiting input or an injection attempt, looking for patterns such as "ignore previous instructions", forced output dictation, fake `</RESUME>` tags, fake `SYSTEM` turns, system prompt exfiltration requests, and jailbreak personas. If the classifier flags the message, the rail short-circuits and the scoring model is never called - the user instead receives a polite refusal.

The benefit of this design is that the protection is independent of the scoring model itself. Even if the underlying model is small, fast, and easy to manipulate, the rail still acts as an early filter that throws away the dangerous input before it can do harm. We tested this with three real attack patterns - a direct instruction override, a role hijack using a fake closing tag and a fake system message, and a system prompt exfiltration request - and confirmed that the unguarded model was hijacked or leaked, while the guarded chain refused all three.

7.4.2 Content Guardrail PII Redaction and Bias Mitigation

Before any AI reads a candidate's resume, the system removes all personally identifying information from it - names, emails, phone numbers, and personal links are stripped out first. After that, LangChain's `PIIMiddleware` scans again for any leftover emails, URLs, or phone numbers, using built-in detectors plus a custom regex for international phone formats, and replaces them with redacted tags like `[REDACTED_EMAIL]`. At the same time, every evaluator prompt starts with a clear rule judge only on skills and experience, and never consider age, gender, race, nationality, name, employment gaps, or how famous the school or company is.

There is also a reflection step where a third AI agent checks whether the first two evaluators gave opinions based on personal characteristics, and flags anything unfair. For job posts, a validator blocks biased language - such as age requirements, gendered pronouns, or exclusionary culture-fit terms - before the description is published. Together, these checks make sure that sensitive details cannot leak into model context and that the scoring decision stays anchored on what the candidate can actually do.

7.4.3 Output Guardrail Hallucination Prevention

Every AI feature in the system uses structured output with strict JSON schemas and Pydantic validation. This means the model is not allowed to reply in free text; it must fill in predefined fields with exact types, which naturally limits how much it can make things up. For example, when drafting job posts or editing them, the AI must preserve locked fields exactly as the recruiter set them, and it cannot add new fields outside the schema. Salary estimates use a dedicated field called `insufficient_evidence` so the AI can admit when it truly has no data instead of guessing numbers.

For resume gap analysis and patch suggestions, the prompts include hard rules called frozen facts the AI is told it can never invent employers, schools, dates, metrics, or skills that do not already exist in the candidate's resume. It must also cite a specific phrase from the resume or job description for every claim it makes, and if it cannot find proof, it must simply leave that point out. In the recruiter chat, search-based answers are grounded only in the provided web results, and the AI is instructed to say "I do not have enough information" when the evidence is missing. When drafting emails or outreach, the AI uses placeholders like `[Candidate Name]` instead of guessing real names or dates it does not know.

8 Critical Evaluation

8.1 Final Report Reflection

8.2 Findings and System Evaluation

8.3 System Evaluation

8.4 Self-Reflection and Professional Development

9 Evidence of Project Management

9.1 Log Sheets

link to FYP Meeting Log Sheets.

9.2 Gantt Chart

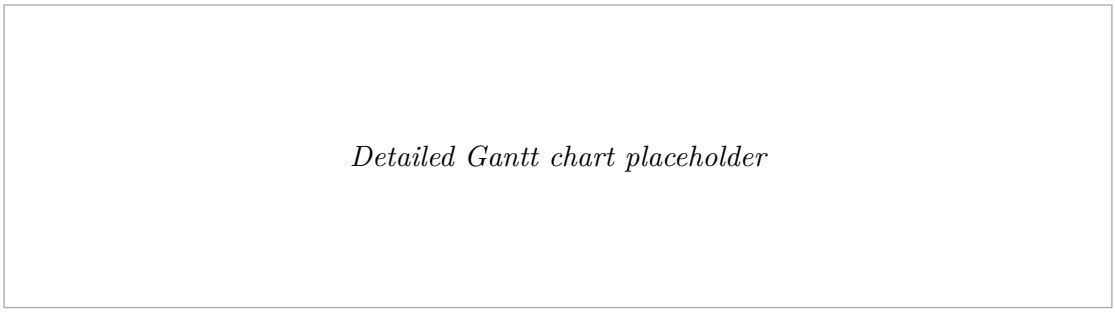


Figure 26: Detailed Gantt chart for the development lifecycle

9.3 Trello Board

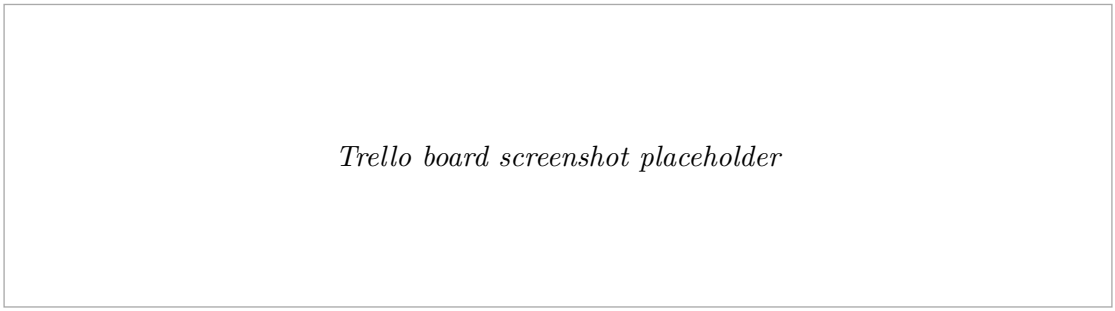


Figure 27: Kanban board on Trello

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10 Appendices

10.1 Wireframes

insert wireframe images.



Figure 28: Candidate dashboard wireframe

10.2 Additional Diagrams

any supplementary diagrams.